

Application of Response Surface Methodology for the Selective Extraction of Lithium Using Hydrophobic Deep Eutectic Solvent

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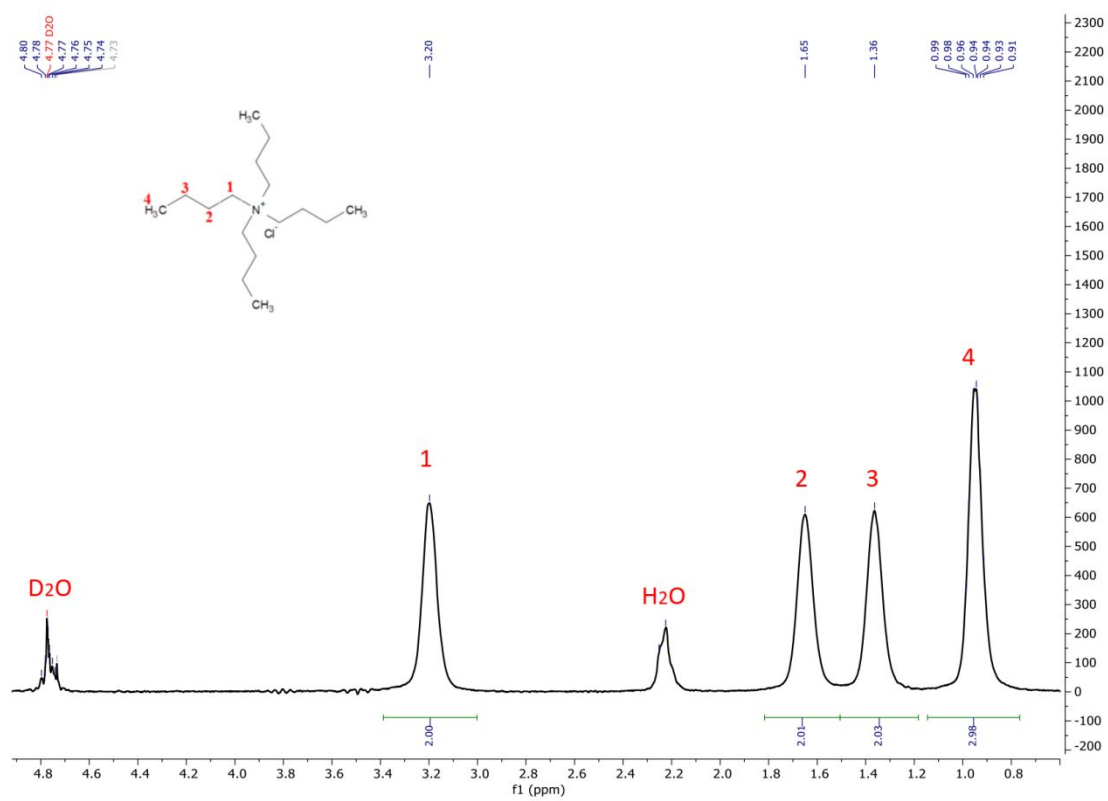


Figure S1. ^1H -NMR spectrum of TBAC.

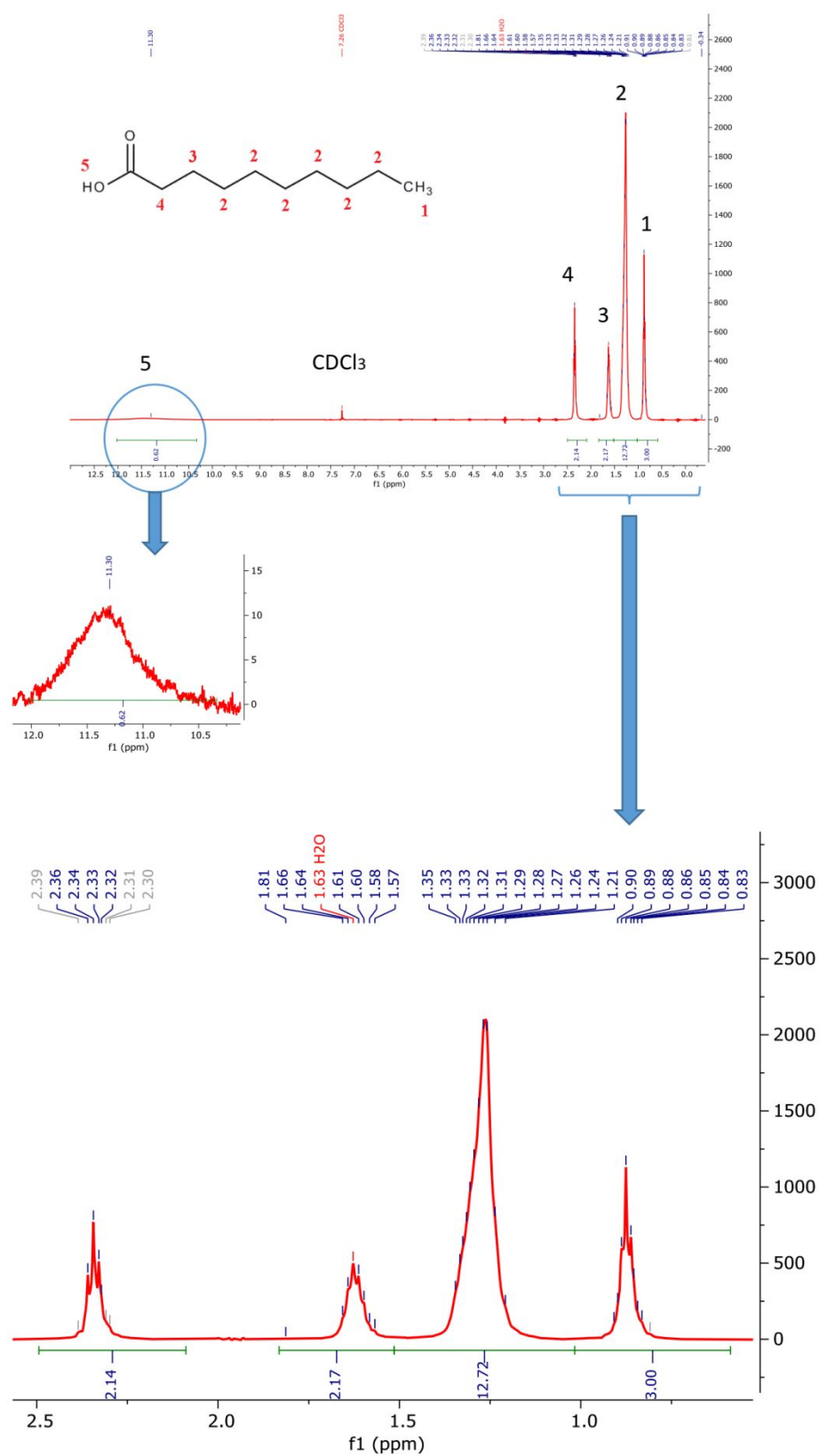


Figure S2. ^1H -NMR spectrum of decanoic acid.

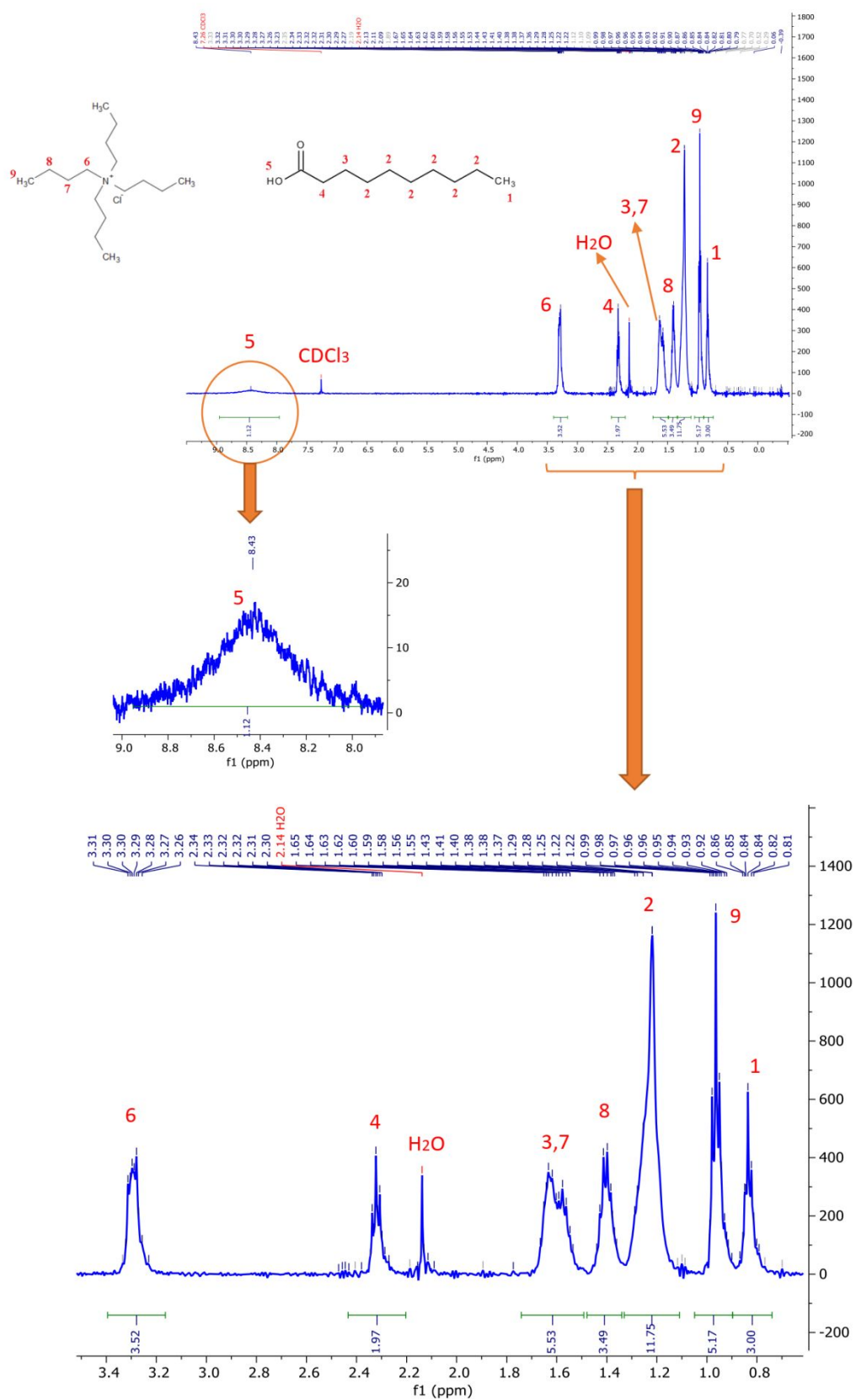


Figure S3. ^1H -NMR spectrum of DES TBAC(1)/DA(2).

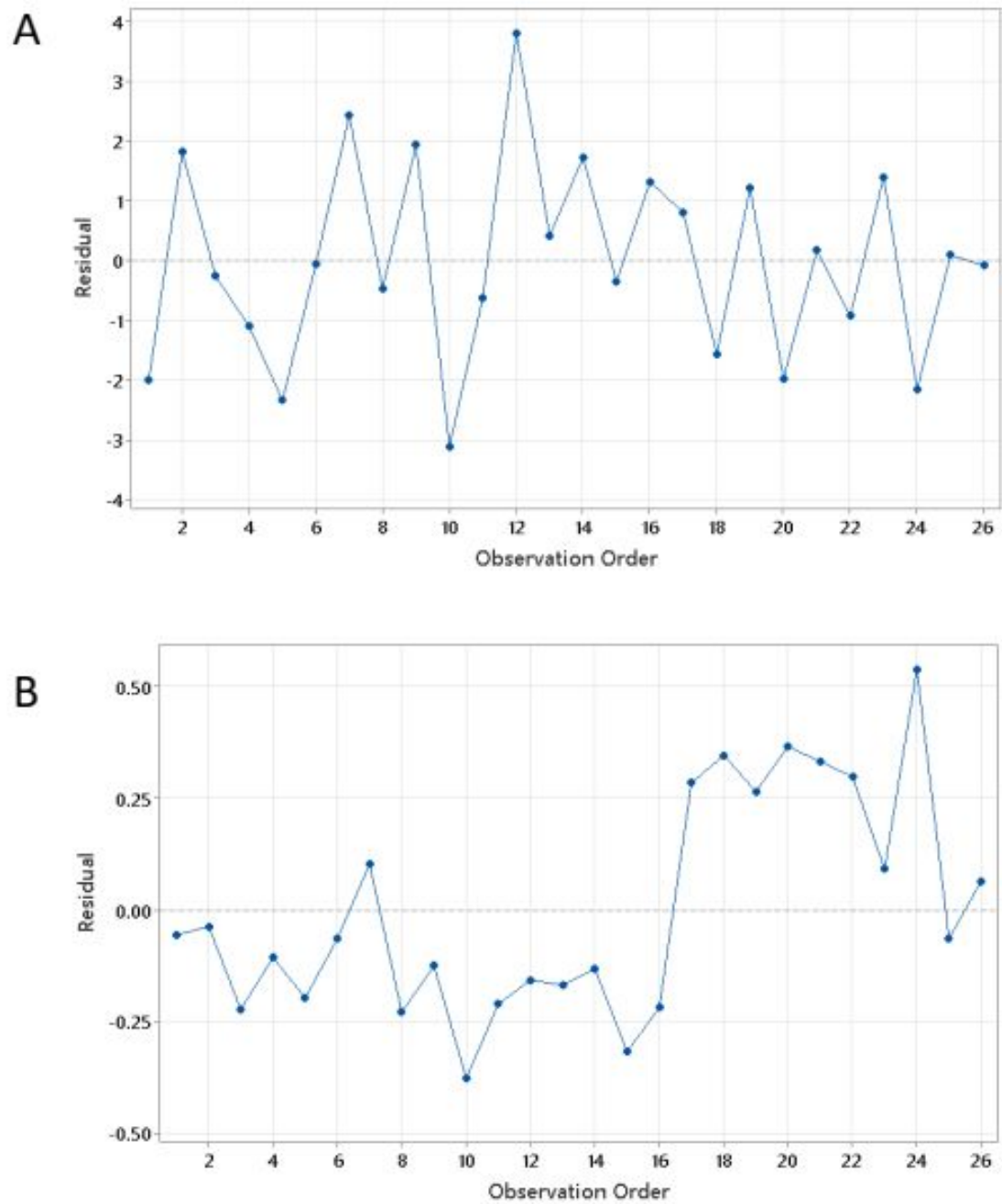


Figure S4. Residual vs. observation order for (A) lithium extraction percentage, (B) selectivity.

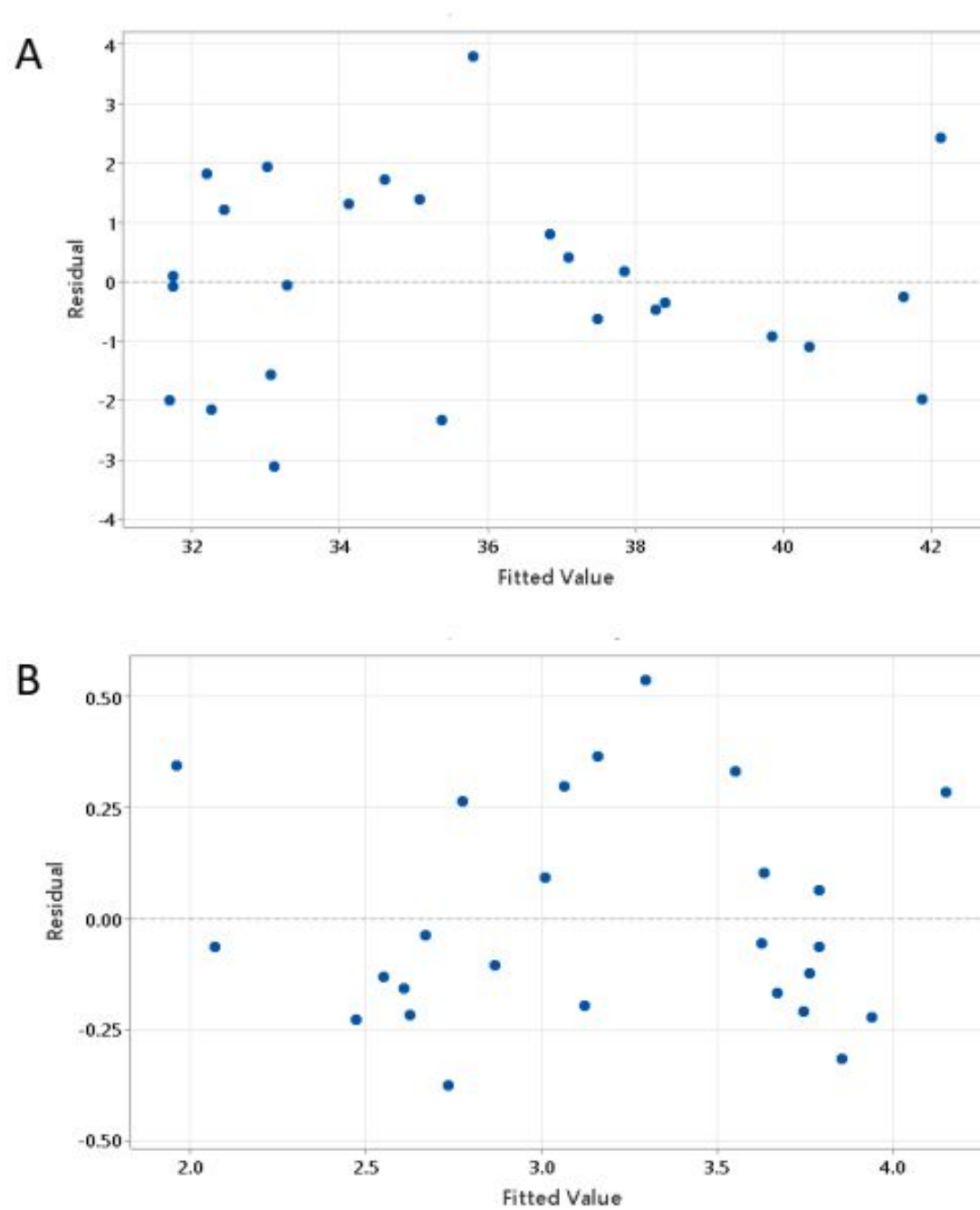


Figure S5. Distribution of residuals. (A) Lithium extraction percentage, (B) Selectivity.

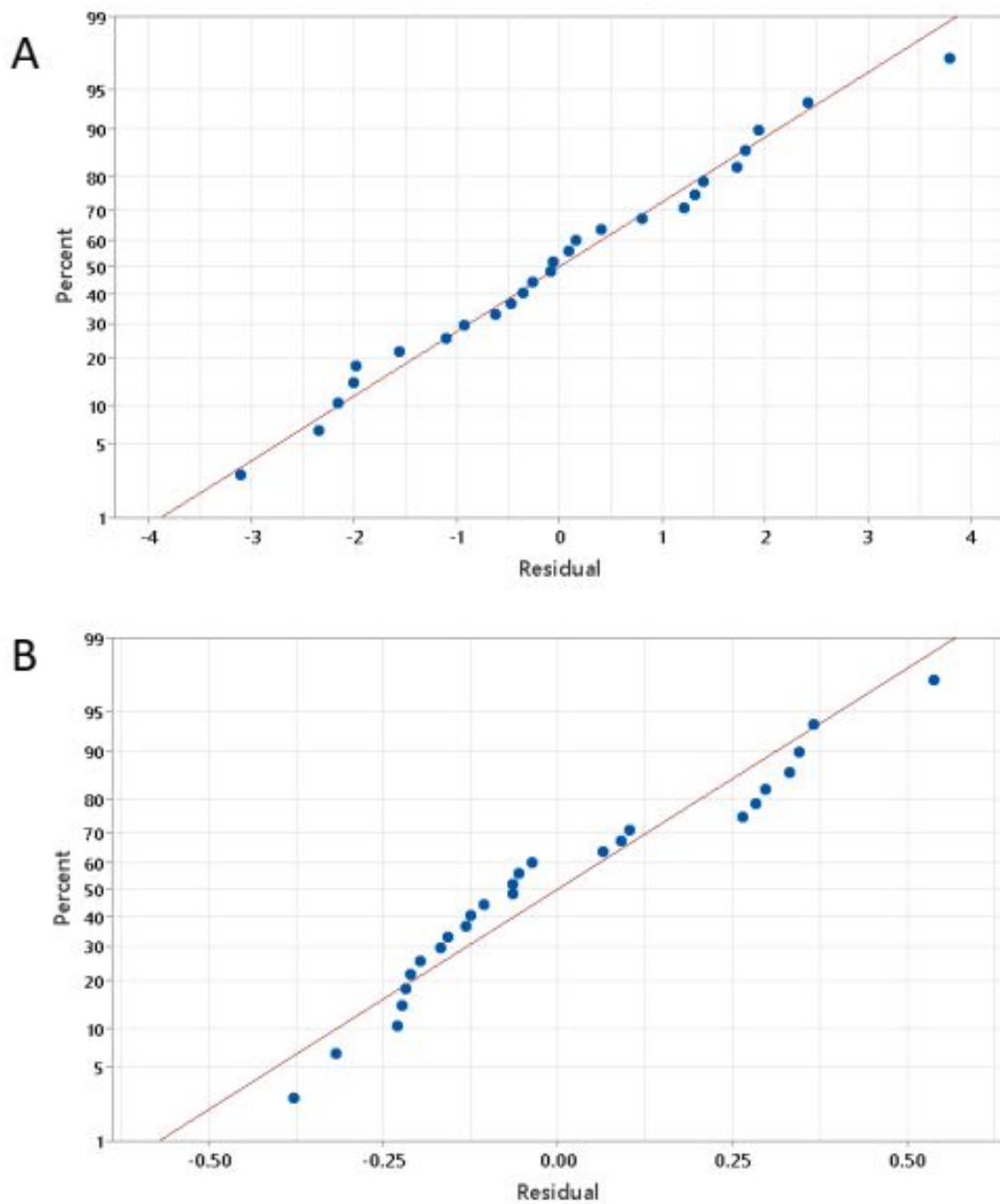


Figure S6. Normal probability plot of residuals. (A) Lithium extraction percentage, (B) Selectivity.

Table S1. Equilibrium mol fraction data for designed experiments (aqueous phase).

No. of exp.	Aqueous phase mol fraction						
	H ₂ O	Li ⁺	Na ⁺	Cl ⁻	H ⁺	OH ⁻	NH ₄ ⁺
1	9.7919E-01	1.9497E-03	6.8984E-03	1.0406E-02	1.5576E-03	5.6968E-07	5.6968E-07
2	9.7919E-01	1.9673E-03	6.9743E-03	1.0406E-02	1.4640E-03	5.6968E-07	5.6968E-07
3	9.7952E-01	2.0898E-03	7.3913E-03	1.0237E-02	7.5558E-04	5.6968E-06	5.6968E-06
4	9.7952E-01	1.9787E-03	7.0332E-03	1.0237E-02	1.2248E-03	5.6968E-06	5.6968E-06
5	9.7919E-01	1.9130E-03	6.7894E-03	1.0406E-02	1.7033E-03	5.6968E-07	5.6968E-07
6	9.7919E-01	1.9536E-03	6.6645E-03	1.0406E-02	1.7875E-03	5.6968E-07	5.6968E-07
7	9.7952E-01	2.0213E-03	6.8816E-03	1.0237E-02	1.3337E-03	5.6968E-06	5.6968E-06
8	9.7952E-01	1.9319E-03	6.5346E-03	1.0237E-02	1.7701E-03	5.6968E-06	5.6968E-06
9	9.5884E-01	2.0008E-03	1.5034E-02	2.0579E-02	3.5447E-03	5.6968E-07	5.6968E-07
10	9.5884E-01	2.0375E-03	1.5410E-02	2.0579E-02	3.1314E-03	5.6968E-07	5.6968E-07
11	9.5900E-01	2.1375E-03	1.6130E-02	2.0494E-02	2.2273E-03	5.6968E-06	5.6968E-06
12	9.5900E-01	2.0482E-03	1.5694E-02	2.0494E-02	2.7518E-03	5.6968E-06	5.6968E-06
13	9.5884E-01	2.0395E-03	1.5427E-02	2.0579E-02	3.1130E-03	5.6968E-07	5.6968E-07
14	9.5884E-01	2.0667E-03	1.5595E-02	2.0579E-02	2.9172E-03	5.6968E-07	5.6968E-07
15	9.5900E-01	2.0530E-03	1.5516E-02	2.0494E-02	2.9250E-03	5.6968E-06	5.6968E-06
16	9.5900E-01	2.0249E-03	1.5280E-02	2.0494E-02	3.1896E-03	5.6968E-06	5.6968E-06
17	9.6952E-01	2.0296E-03	1.1328E-02	1.5237E-02	1.8802E-03	1.8015E-06	1.8015E-06
18	9.6952E-01	2.0134E-03	1.1303E-02	1.5237E-02	1.9213E-03	1.8015E-06	1.8015E-06
19	9.6880E-01	2.0272E-03	1.1336E-02	1.5601E-02	2.2377E-03	1.8015E-07	1.8015E-07
20	9.6968E-01	2.0387E-03	1.1317E-02	1.5143E-02	1.7878E-03	1.8015E-05	1.8015E-05
21	9.6952E-01	2.0679E-03	1.1698E-02	1.5237E-02	1.4715E-03	1.8015E-06	1.8015E-06
22	9.6952E-01	1.9541E-03	1.1046E-02	1.5237E-02	2.2372E-03	1.8015E-06	1.8015E-06
23	9.8659E-01	2.2630E-03	3.8203E-03	6.7021E-03	6.1880E-04	1.8015E-06	1.8015E-06
24	9.5141E-01	2.0354E-03	2.0325E-02	2.4292E-02	1.9316E-03	1.8015E-06	1.8015E-06
25	9.6952E-01	2.0611E-03	1.1654E-02	1.5237E-02	1.5227E-03	1.8015E-06	1.8015E-06
26	9.7919E-01	1.9497E-03	6.8984E-03	1.0406E-02	1.5576E-03	5.6968E-07	5.6968E-07

Table S2. Equilibrium mol fraction data for designed experiments (organic phase).

No. of exp.	Aqueous phase mol fraction			
	DES	TOPO	RLi	RNa
1	8.6130E-01	1.1880E-01	4.7236E-03	1.5176E-02
2	8.6760E-01	1.1393E-01	4.4427E-03	1.4028E-02
3	8.7202E-01	1.1853E-01	2.1288E-03	7.3190E-03
4	8.6756E-01	1.1689E-01	3.5709E-03	1.1974E-02
5	7.1477E-01	2.6106E-01	5.7661E-03	1.8399E-02
6	7.1552E-01	2.5954E-01	5.1035E-03	1.9837E-02
7	7.1472E-01	2.6607E-01	3.4381E-03	1.5772E-02
8	7.1844E-01	2.5706E-01	4.5411E-03	1.9958E-02
9	8.4176E-01	1.1369E-01	5.1180E-03	3.9428E-02
10	8.4358E-01	1.1641E-01	4.7347E-03	3.5273E-02
11	8.5388E-01	1.1789E-01	3.1148E-03	2.5112E-02
12	8.4718E-01	1.1782E-01	4.2631E-03	3.0743E-02
13	6.9451E-01	2.6151E-01	5.2067E-03	3.8769E-02
14	6.9744E-01	2.6138E-01	4.8193E-03	3.6357E-02
15	6.9708E-01	2.6097E-01	4.7372E-03	3.7215E-02
16	6.9159E-01	2.6290E-01	5.1138E-03	4.0400E-02
17	7.8712E-01	1.8740E-01	4.1829E-03	2.1296E-02
18	7.8816E-01	1.8597E-01	4.3741E-03	2.1494E-02
19	7.8325E-01	1.8704E-01	4.4430E-03	2.5266E-02
20	7.9038E-01	1.8567E-01	3.6236E-03	2.0326E-02
21	9.2684E-01	5.5569E-02	3.2317E-03	1.4359E-02
22	6.1815E-01	3.4832E-01	5.7578E-03	2.7772E-02
23	8.0493E-01	1.8683E-01	3.6684E-03	4.5749E-03
24	7.8734E-01	1.8697E-01	3.0980E-03	2.2593E-02
25	7.9418E-01	1.8547E-01	3.7028E-03	1.6645E-02
26	8.6130E-01	1.1880E-01	4.7236E-03	1.5176E-02

Table S3. Analysis of variance of lithium extraction data.

Source	DF	Seq SS	Contribution	Adj SS	Adj MS	F-Value	P-Value
Model	14	290.453	80.76%	290.453	20.747	3.30	0.027
Linear	4	172.802	48.05%	172.802	43.201	6.87	0.005
A	1	21.271	5.91%	21.271	21.271	3.38	0.093
B	1	133.643	37.16%	133.643	133.643	21.25	0.001
C	1	5.966	1.66%	5.966	5.966	0.95	0.351
D	1	11.923	3.32%	11.923	11.923	1.90	0.196
Square	4	67.732	18.83%	67.732	16.933	2.69	0.087
A ²	1	0.084	0.02%	11.260	11.260	1.79	0.208
B ²	1	10.242	2.85%	31.992	31.992	5.09	0.045
C ²	1	53.352	14.83%	55.076	55.076	8.76	0.013
D ²	1	4.054	1.13%	4.054	4.054	0.64	0.439
2-Way Interaction	6	49.919	13.88%	49.919	8.320	1.32	0.325
AB	1	3.138	0.87%	3.138	3.138	0.50	0.495
AC	1	6.668	1.85%	6.668	6.668	1.06	0.325
AD	1	0.166	0.05%	0.166	0.166	0.03	0.874
BC	1	10.030	2.79%	10.030	10.030	1.59	0.233
BD	1	29.752	8.27%	29.752	29.752	4.73	0.052
CD	1	0.165	0.05%	0.165	0.165	0.03	0.874

Table S4. Analysis of variance of selectivity data.

Source	DF	Seq SS	Contribution	Adj SS	Adj MS	F-Value	P-Value
Model	14	9.1554	85.92%	9.15543	0.65396	4.80	0.006
Linear	4	7.8639	73.80%	7.86390	1.96597	14.42	0.000
A	1	7.1635	67.23%	7.16354	7.16354	52.53	0.000
B	1	0.2197	2.06%	0.21969	0.21969	1.61	0.231
C	1	0.3574	3.35%	0.35736	0.35736	2.62	0.134
D	1	0.1233	1.16%	0.12330	0.12330	0.90	0.362
Square	4	0.9493	8.91%	0.94926	0.23732	1.74	0.211
A ²	1	0.1192	1.12%	0.58572	0.58572	4.29	0.063
B ²	1	0.3392	3.18%	0.73448	0.73448	5.39	0.041
C ²	1	0.0507	0.48%	0.25211	0.25211	1.85	0.201
D ²	1	0.4401	4.13%	0.44012	0.44012	3.23	0.100
2-Way Interaction	6	0.3423	3.21%	0.34227	0.05705	0.42	0.852
AB	1	0.0122	0.11%	0.01220	0.01220	0.09	0.770
AC	1	0.0082	0.08%	0.00821	0.00821	0.06	0.811
AD	1	0.0047	0.04%	0.00473	0.00473	0.03	0.856
BC	1	0.0400	0.38%	0.03999	0.03999	0.29	0.599
BD	1	0.1067	1.00%	0.10675	0.10675	0.78	0.395
CD	1	0.1704	1.60%	0.17040	0.17040	1.25	0.287

